

# FRANCK JUNIOR ABOYA MESSOU

AI Researcher & Engineer | LLM / sLM for Smart Grids & IoT | Multi-GPU Infrastructure

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## SUMMARY

PhD researcher and AI engineer working at the intersection of **small / large language models (sLM / LLM)**, time-series forecasting, and IoT-instrumented power systems. First-author of **2 Q1 journal papers** (*Internet of Things*, *Expert Systems with Applications*) and **5 IEEE flagship conference papers** (VTC, GLOBECOM, ICC, ICCE). Build and operate **dedicated multi-GPU servers** end-to-end: CUDA toolchain, distributed training, model checkpointing, and inference serving. Ship LLM-powered production prototypes for industry clients through JICA’s consulting practice.

## EXPERIENCE

ML Data Engineer (Part-time)

D’isum Inc.

 Oct 2025 – Jun 2026  Tokyo, Japan

- Built **IoT sensor-data pipelines** from raw streams to analytics-ready datasets feeding ML prototypes.
- Integrated **ML / predictive models** into client systems; precision-sensor analytics and system fusion.
- Sensor apps (iOS & Android)**: built with Java and Swift.

Doctoral Researcher

**YuLab, Hosei University** (Advisor: Prof. Keping Yu, Stanford / Elsevier World’s Top 2% Scientists 2025)

 Oct 2025 – Present  Tokyo, Japan

- Design dual-domain **graph-neural-network architectures** for power-system stability at PEGASE-2,869-bus scale.
- Develop the **spectral-to-language reprogramming** pipeline (Spec2Llama, Spec2LLM, LLM4Imp) targeting both LLM and **small-language-model** backbones for IoT-edge deployment.
- Provision and operate the **lab multi-GPU server**: CUDA / PyTorch, distributed training (DDP), model checkpointing, inference services.

Master’s Researcher

**YuLab, Hosei University**

 2023 – Sep 2025  Tokyo, Japan

- Foundational work on **attention-based transformers** for IoT load forecasting: TAPformer, TSFormer, hybrid CNN-BiLSTM.
- Set up the lab GPU-server training pipeline that the doctoral work now extends.
- Awarded **IEEE VTS Tokyo / Japan Chapter 2024 Student Paper Award** at VTC2024-Fall (Washington D.C.).

## EDUCATION

PhD, Applied Informatics

**Hosei University**

 2023 – Apr 2027 (early)  Tokyo, Japan

Thesis: AI / sLM / LLM methods for smart-grid stability, multi-impairment analysis, and SCADA anomaly detection on IoT-instrumented power systems.

MSc, Applied Informatics

**Hosei University**

 2023 – Sep 2025  Tokyo, Japan

Transformer-based forecasting for IoT energy systems (TAPformer, TSFormer, CNN-BiLSTM hybrid).

## TECHNICAL SKILLS

AI / ML & LLMs

PyTorch Hugging Face vLLM scikit-learn  
OpenAI / GPT Anthropic / Claude  
Llama Qwen Phi SmolLM

Research methods

Time-series forecasting Spectral reprogramming  
Graph neural networks Federated learning  
Prompt engineering Adversarial testing

Multi-GPU infrastructure

CUDA toolchain NVIDIA drivers DDP training  
Inference serving Linux server admin

ML backend & APIs

FastAPI Flask REST Model serving  
Python Java Bash

Data & DevOps

pandas NumPy PostgreSQL MongoDB  
Docker GitHub Actions GCP

Intern, GenAI for Business · TICAD9 Booth

Japan AI Consulting (J-AIC)

📅 Jul 2024 · Aug 2025      📍 Osaka · Yokohama

- Internship: discovery, prototyping, and demonstrating **LLM-powered solutions** for the consulting practice.
- TICAD9: presented ML and GenAI prototypes at JICA’s TICAD9 booth (Yokohama).

PUBLICATIONS

- **MFJ Aboya** et al., “TAPformer: Long-Term Electricity Load Forecasting in IoT Systems,” *Internet of Things*, Elsevier, 2026. (IF 7.6, Q1)
- **MFJ Aboya** et al., “Spec2Llama: Spectral-Aware Forecasting for Versatile Time Series,” *Expert Systems with Applications*, 2026. (IF 7.5, Q1)
- **MFJ Aboya** et al., “Hybrid Attention-Based CNN-BiLSTM for Short-Term Load Forecasting in IoT,” *IEEE VTC2024-Fall*, Washington D.C. **Student Paper Award**
- **MFJ Aboya** et al., “Spec2LLM: Spectral-to-Language Reprogramming for Power Transformer Temperature Forecasting,” *IEEE ICCE 2026*, Dubai.
- **MFJ Aboya** et al., “LLM4Imp: Frozen LLMs with Spectral Prompts for Time-Series Imputation,” *IEEE ICC 2026*, Glasgow.

AI PROJECTS

- **Smart-grid co-optimization:** dual-domain GNN architectures for power-system stability at PEGASE-2,869-bus scale.
- **Spec2Llama / Spec2LLM / LLM4Imp:** spectral-to-language reprogramming for LLM and sLM backbones on IoT-edge devices.
- **Multi-GPU lab infrastructure:** full lifecycle ownership (hardware, CUDA, drivers, DDP, model serving); pattern reused across engineering side-projects.

Public code: [github.com/mesabo](https://github.com/mesabo)

AWARDS & SERVICE

-  **IEEE VTS Tokyo / Japan Chapter**  
Student Paper Award & Encouragement Award, VTC2024-Fall (Oct 2024)
-  **JICA Development Studies Program**  
Graduate, 2025
-  **Review Committee, IEEE ICECET 2026**  
Rome, IEEE Italy Section
-  **Reviewer, Theoretical Computer Science**  
Elsevier (ISSN 0304-3975)

LANGUAGES

French      ● ● ● ● ●      English      ● ● ● ● ●      Japanese      ● ● ● ● ●